US 20250276824A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0276824 A1
(43) Pub. Date: Sep. 4, 2025
KIM

(54) **PACKAGING BOX**

(71) Applicant: **Sung Min KIM**, Jeonju-si, Jeollabuk-do (KR)

(72) Inventor: **Sung Min KIM**, Jeonju-si, Jeollabuk-do (KR)

(21) Appl. No.: **18/862,570**

(22) PCT Filed: **May 9, 2023**

(86) PCT No.: **PCT/KR2023/006234**

§ 371 (c)(1),
(2) Date: **Nov. 4, 2024**

(30) **Foreign Application Priority Data**

May 13, 2022 (KR) 10-2022-0058922

Publication Classification

(51) **Int. Cl.**
B65D 5/10 (2006.01)
B65D 5/66 (2006.01)

(52) **U.S. Cl.**
CPC *B65D 5/106* (2013.01); *B65D 5/6608* (2013.01)

(57) **ABSTRACT**

The present invention relates to a packaging box, of which a cubic body can be assembled without the use of packaging tape, twine, or separate portable holder clips, wherein the cubic body, after being completely assembled, can be used in holding and easily transporting various kinds of heavy objects such as daily necessities, without the concern of the box being untied from a fastened state by heavy loads.

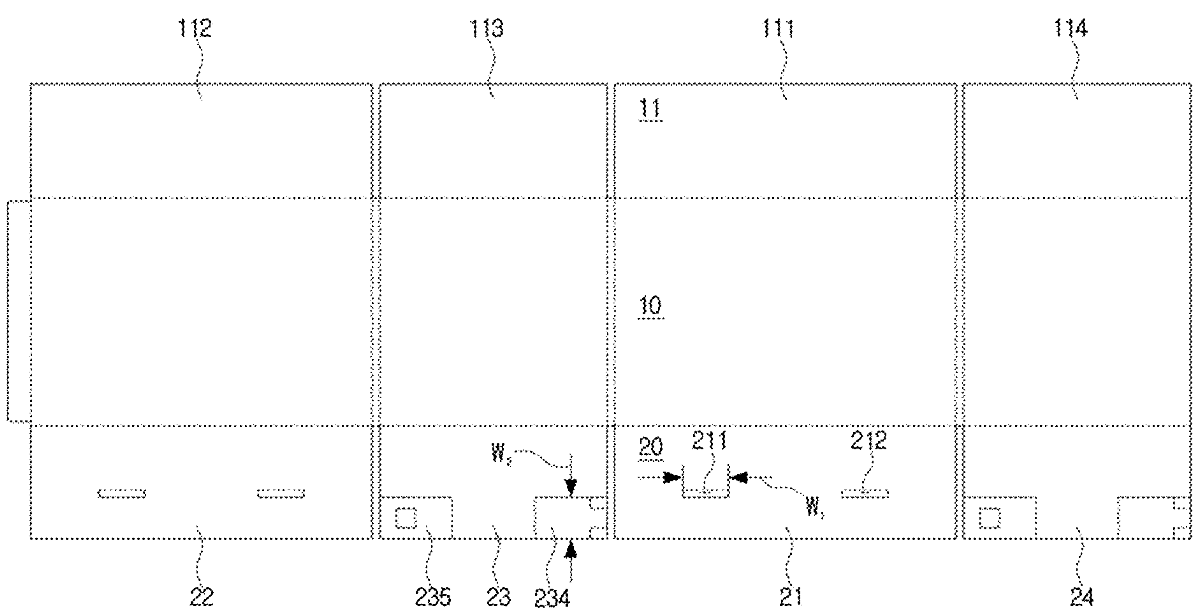


FIG. 1

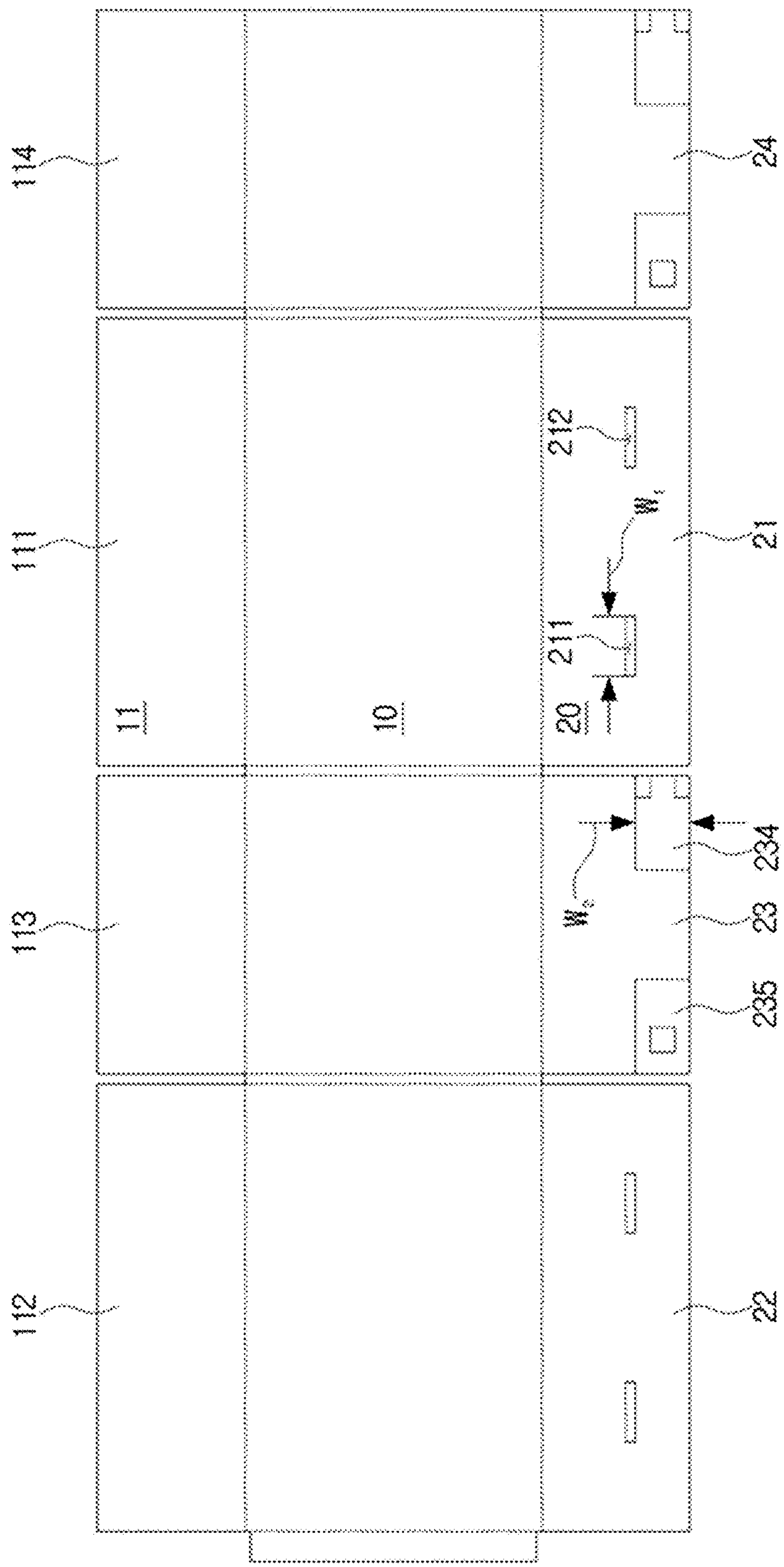


FIG. 2

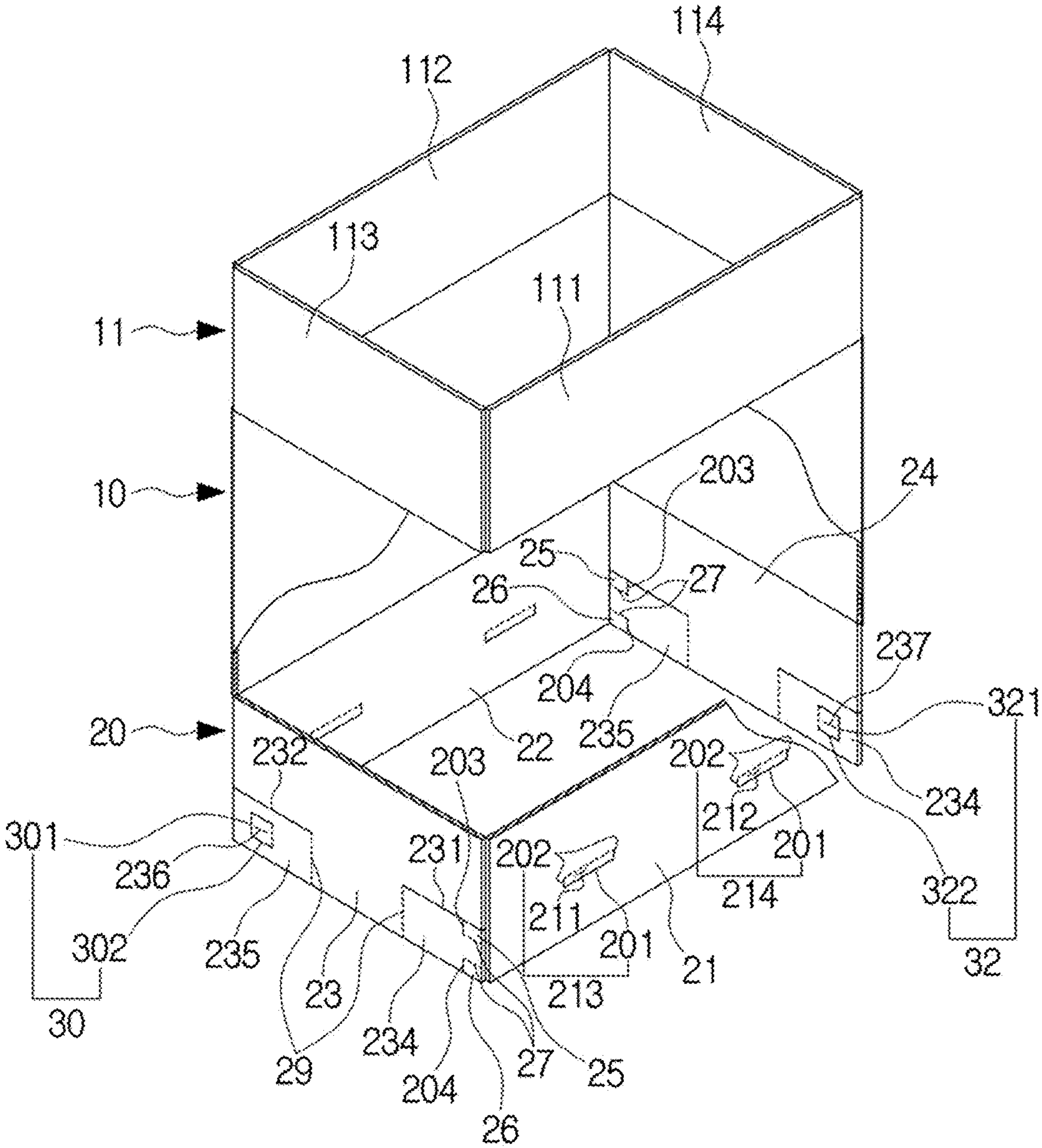

114
112
113
111
11
10
203
24
25
26
27
237
204
235
20
232
203
22
321
202
212
234
301
201
236
231
202
214
302
211
201
322
235
23
213
21
32
30
234
29
27
204
25
26

FIG. 3

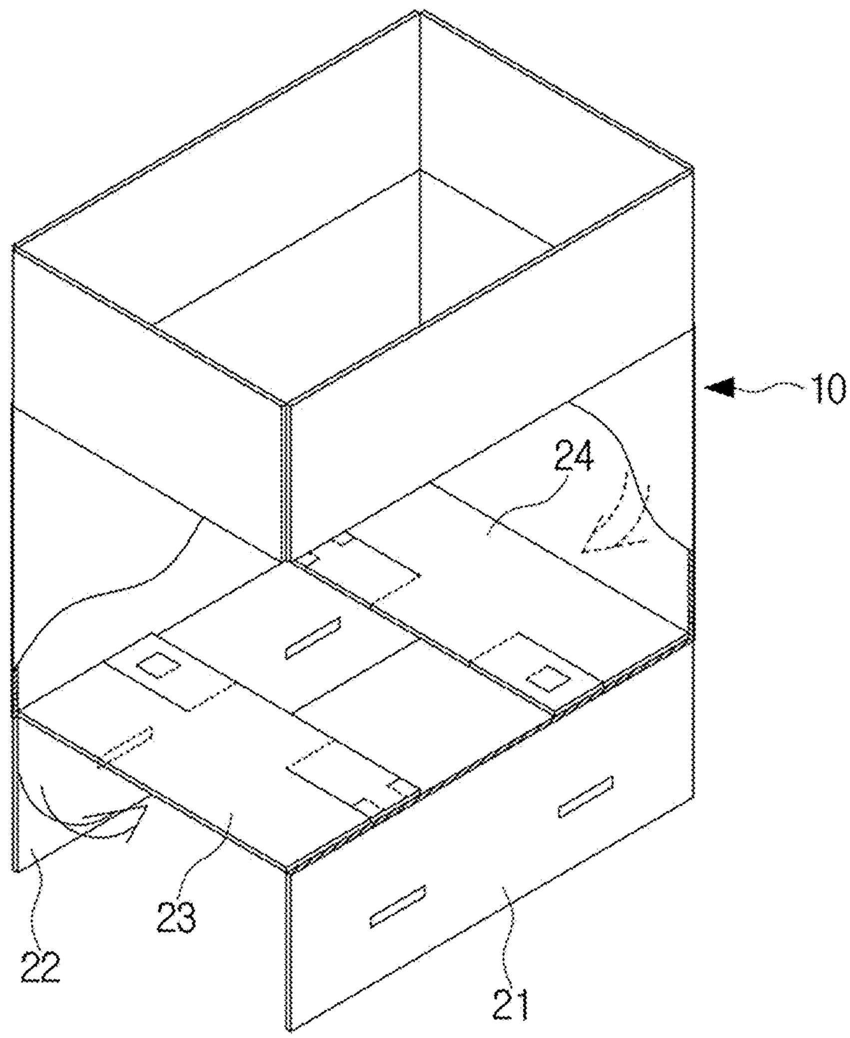


FIG. 4

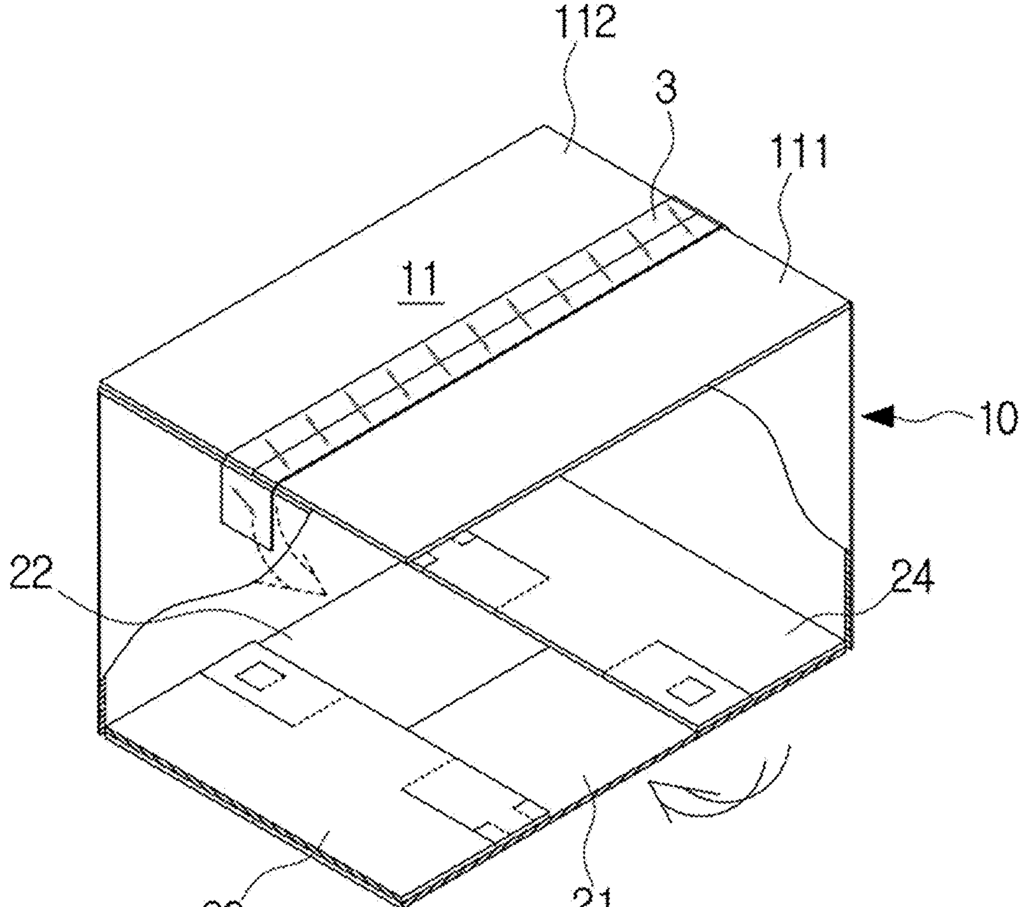


FIG. 5

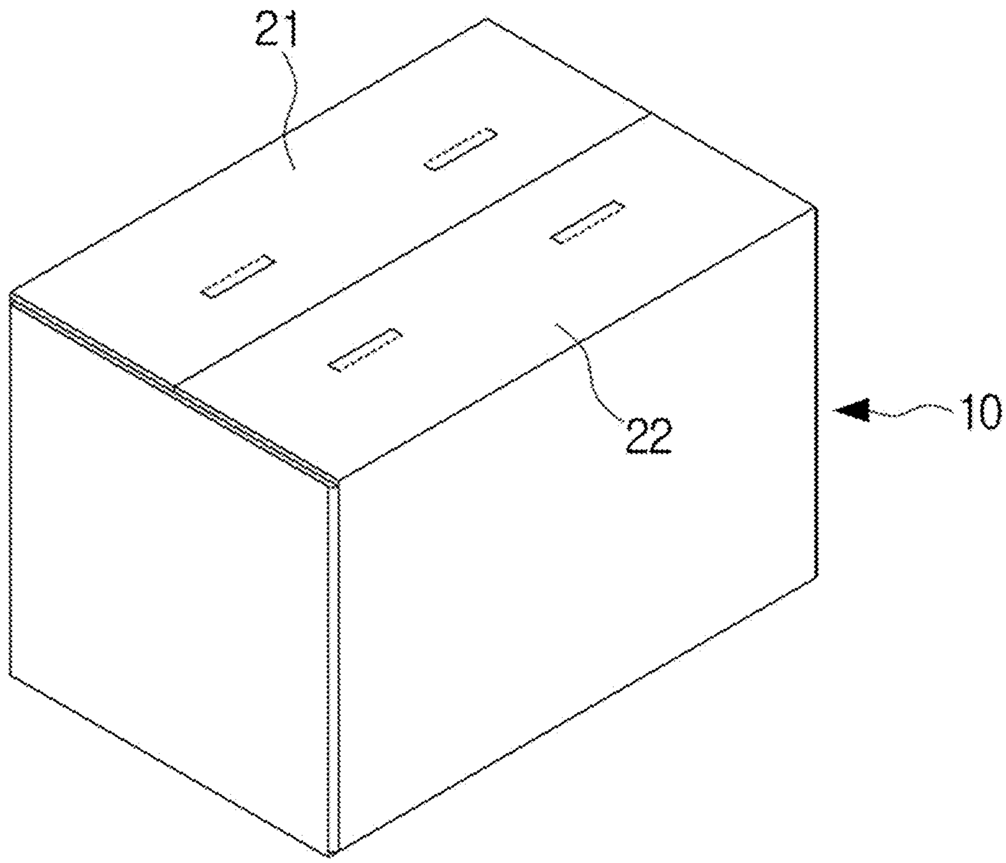


FIG. 6

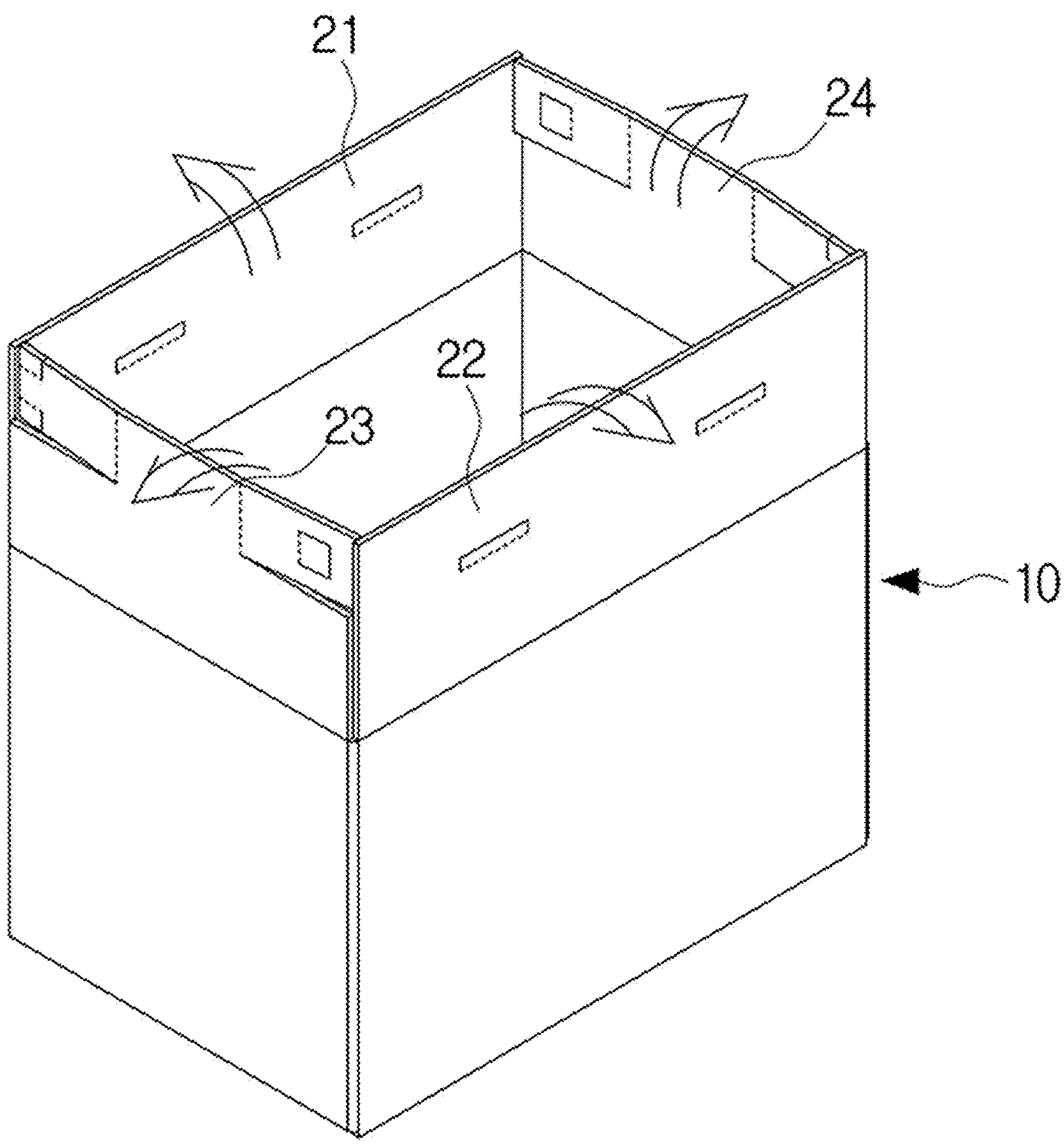


FIG. 7

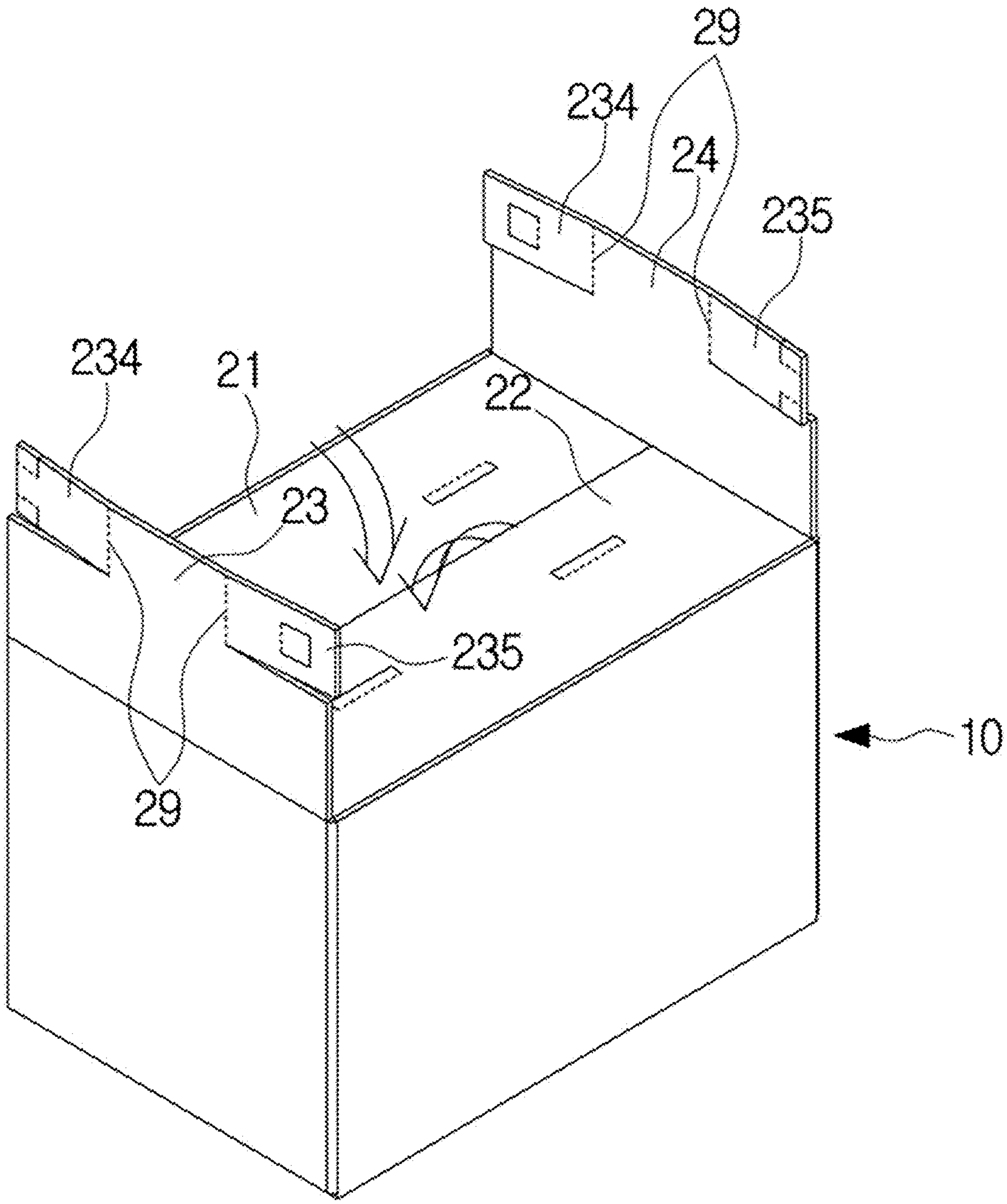


FIG. 8

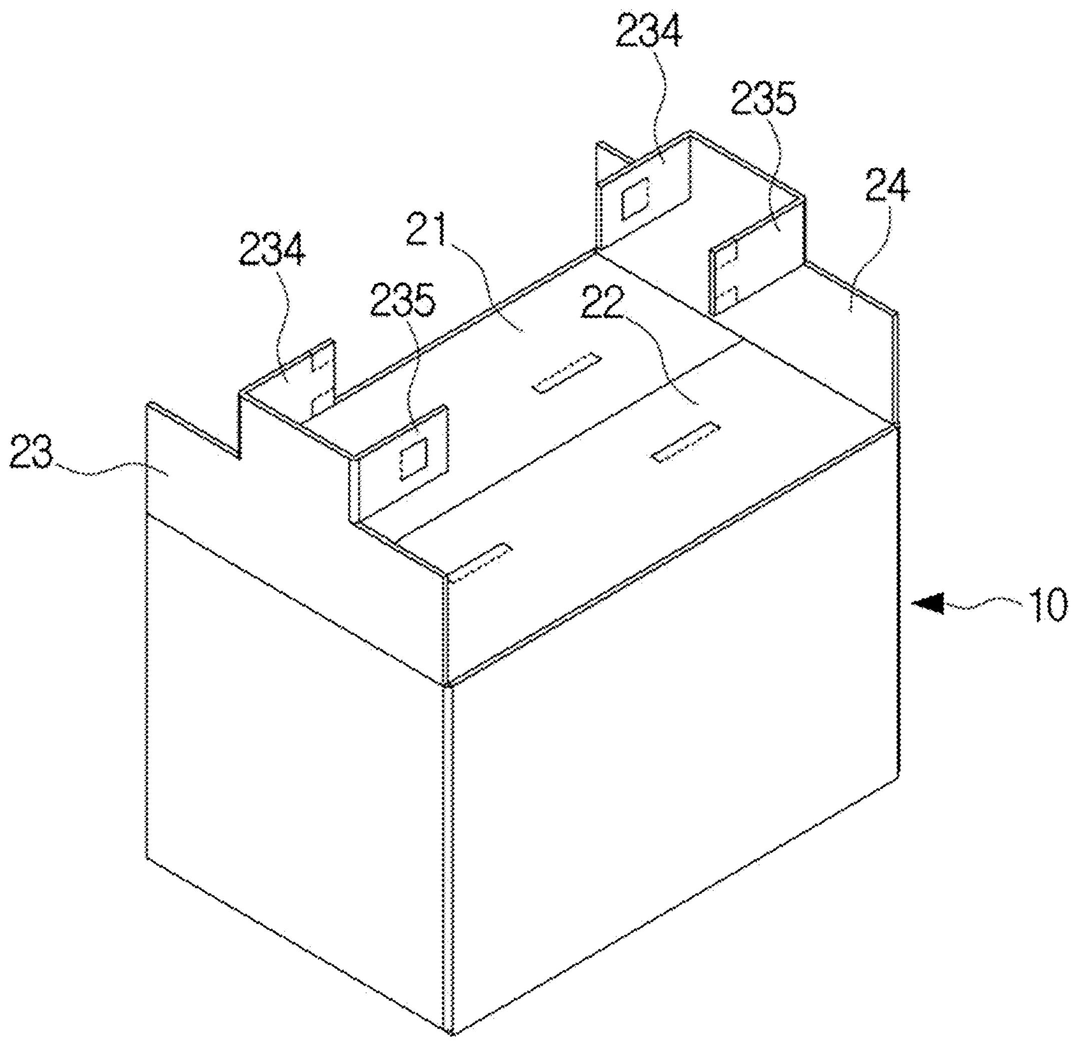


FIG. 9

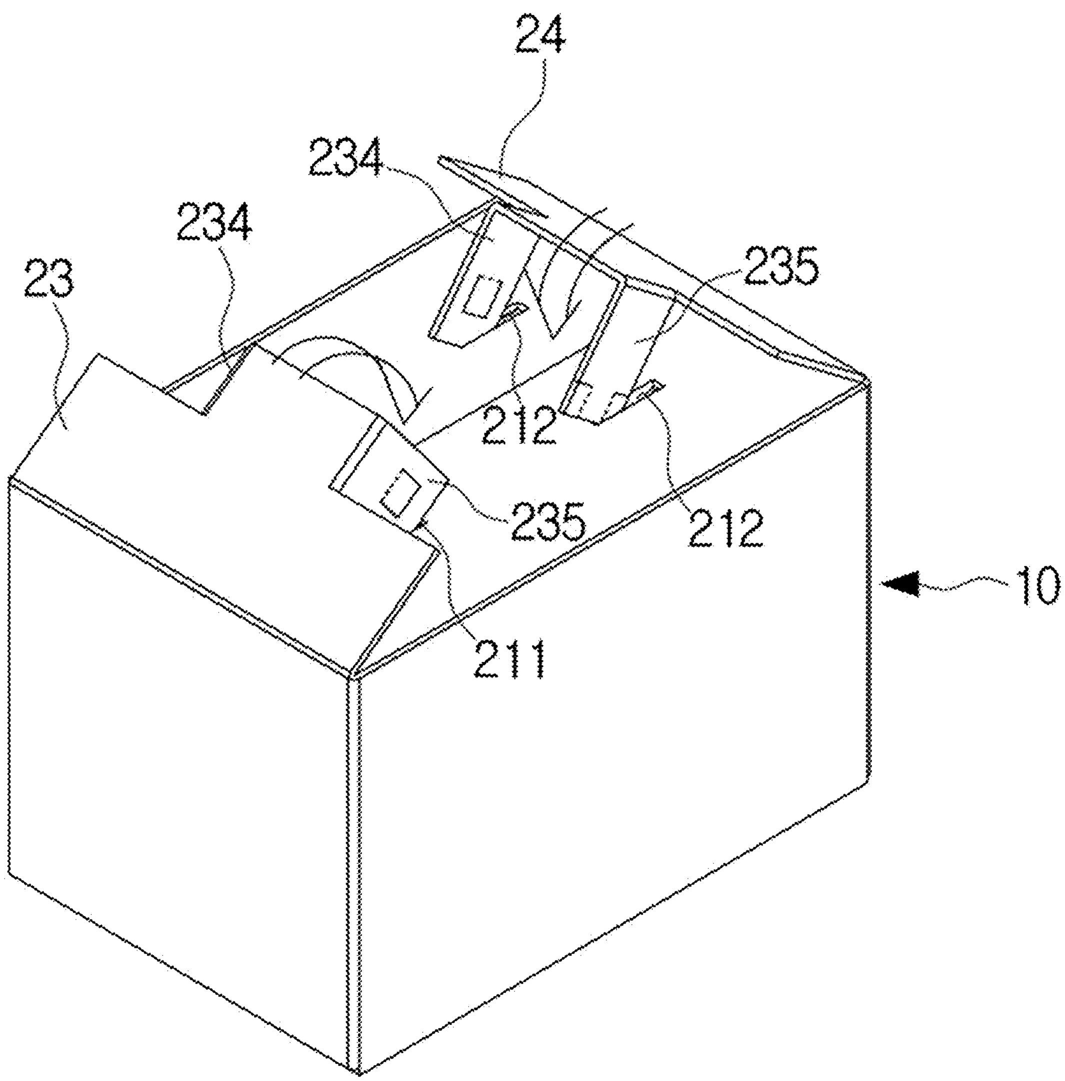


FIG. 10

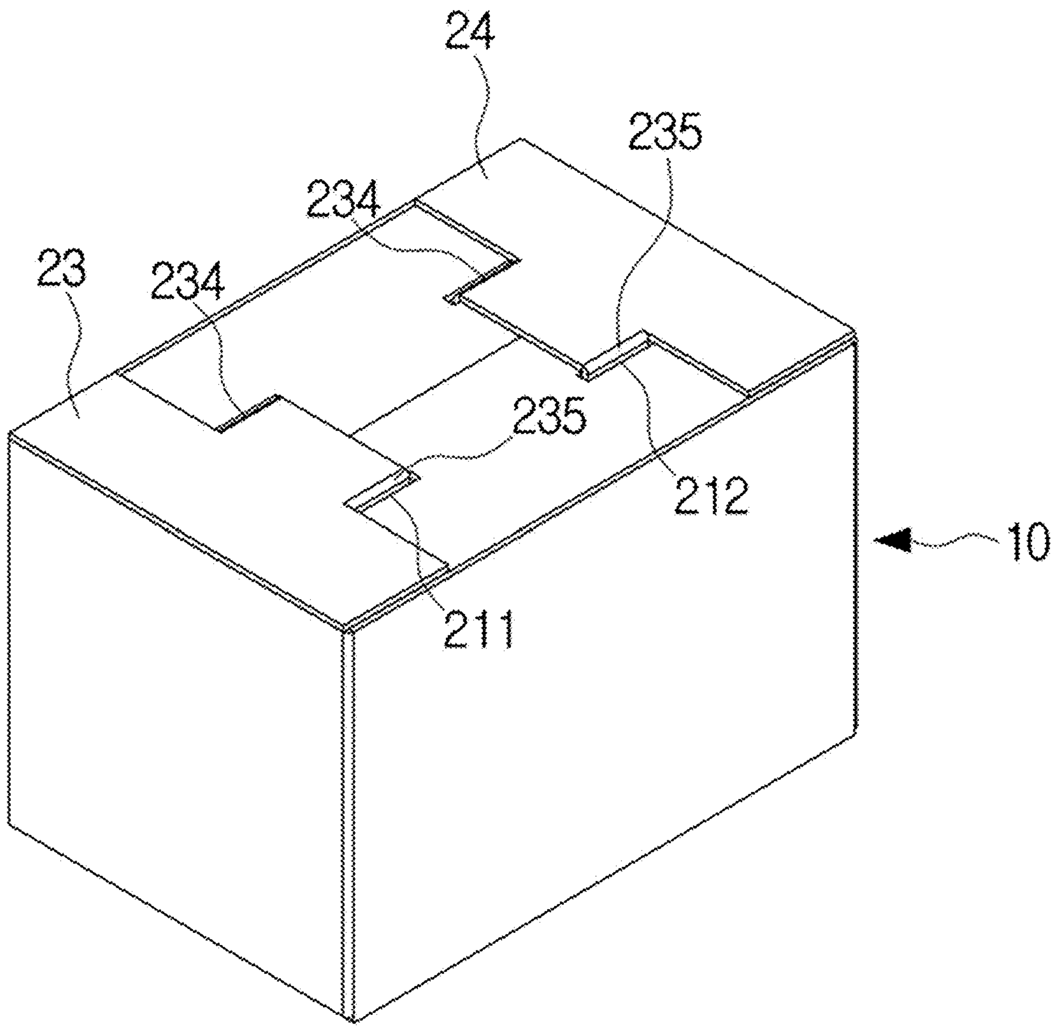


FIG. 11

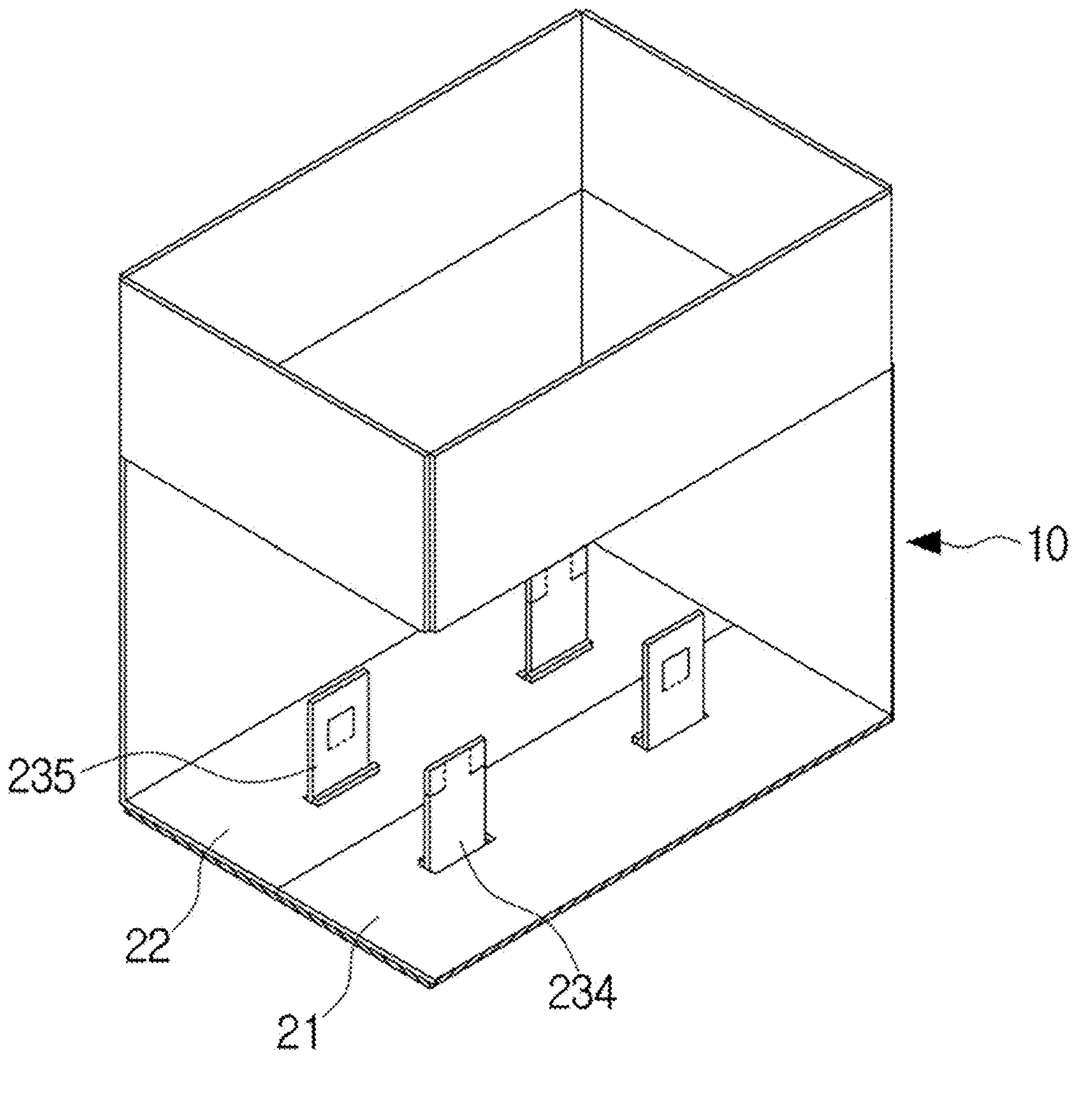


FIG. 12

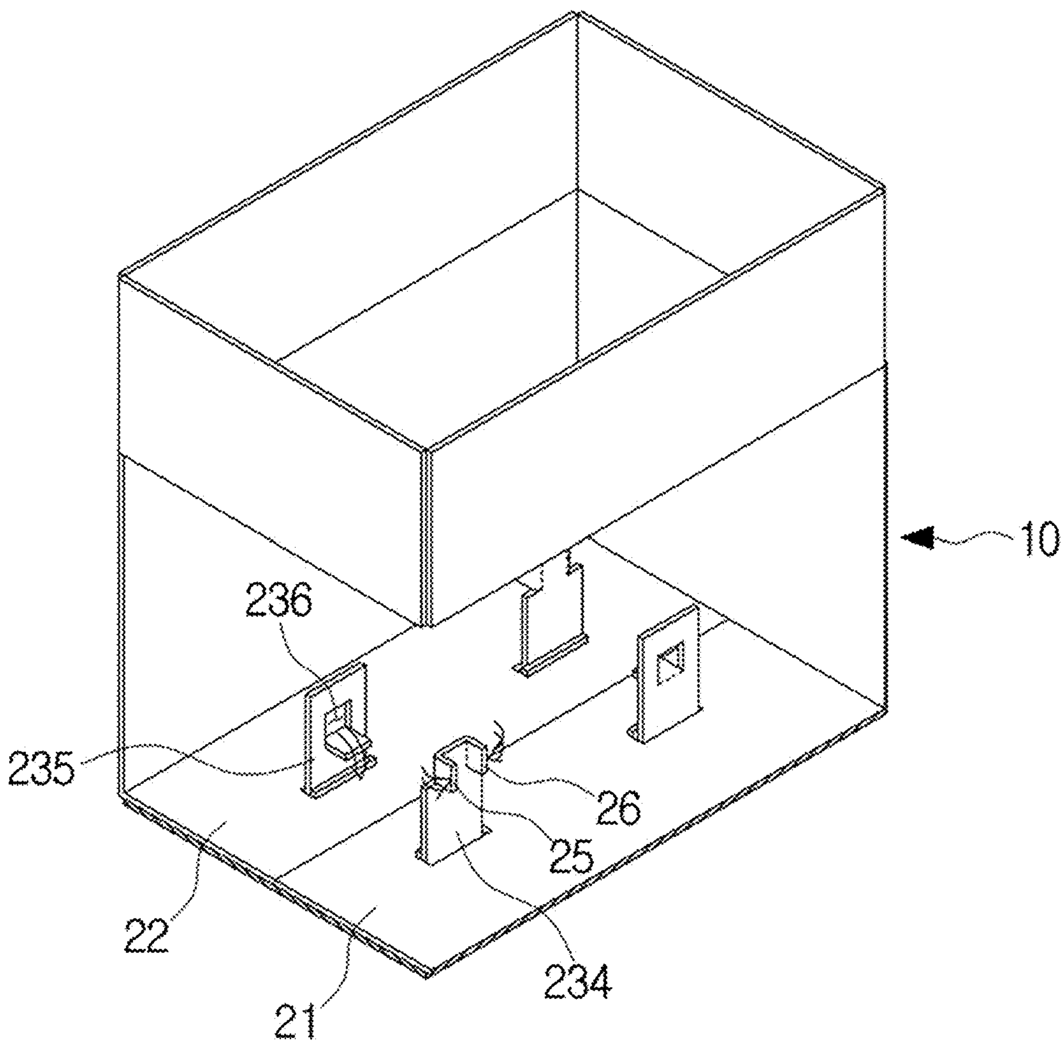


FIG. 13

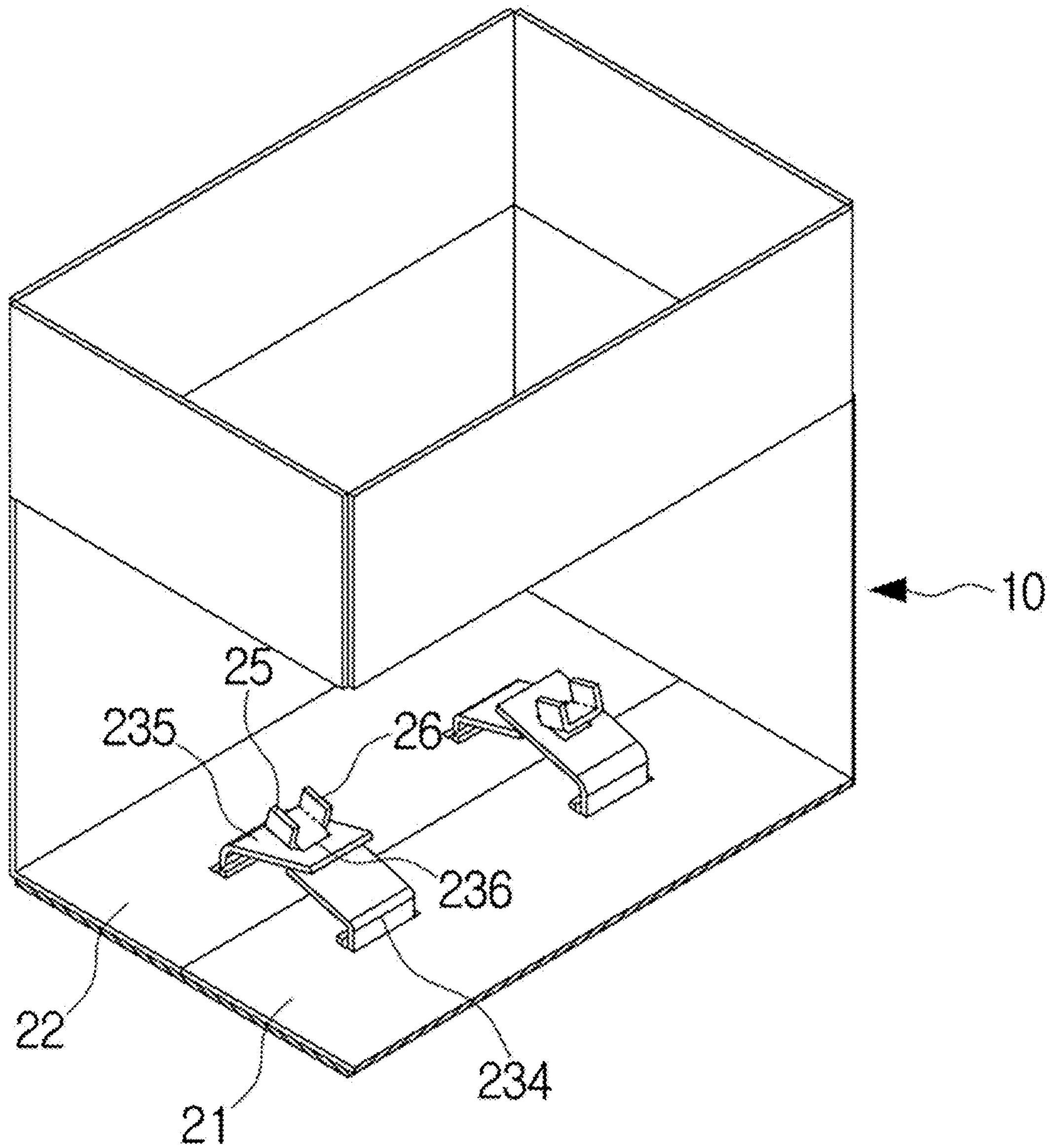


FIG. 14

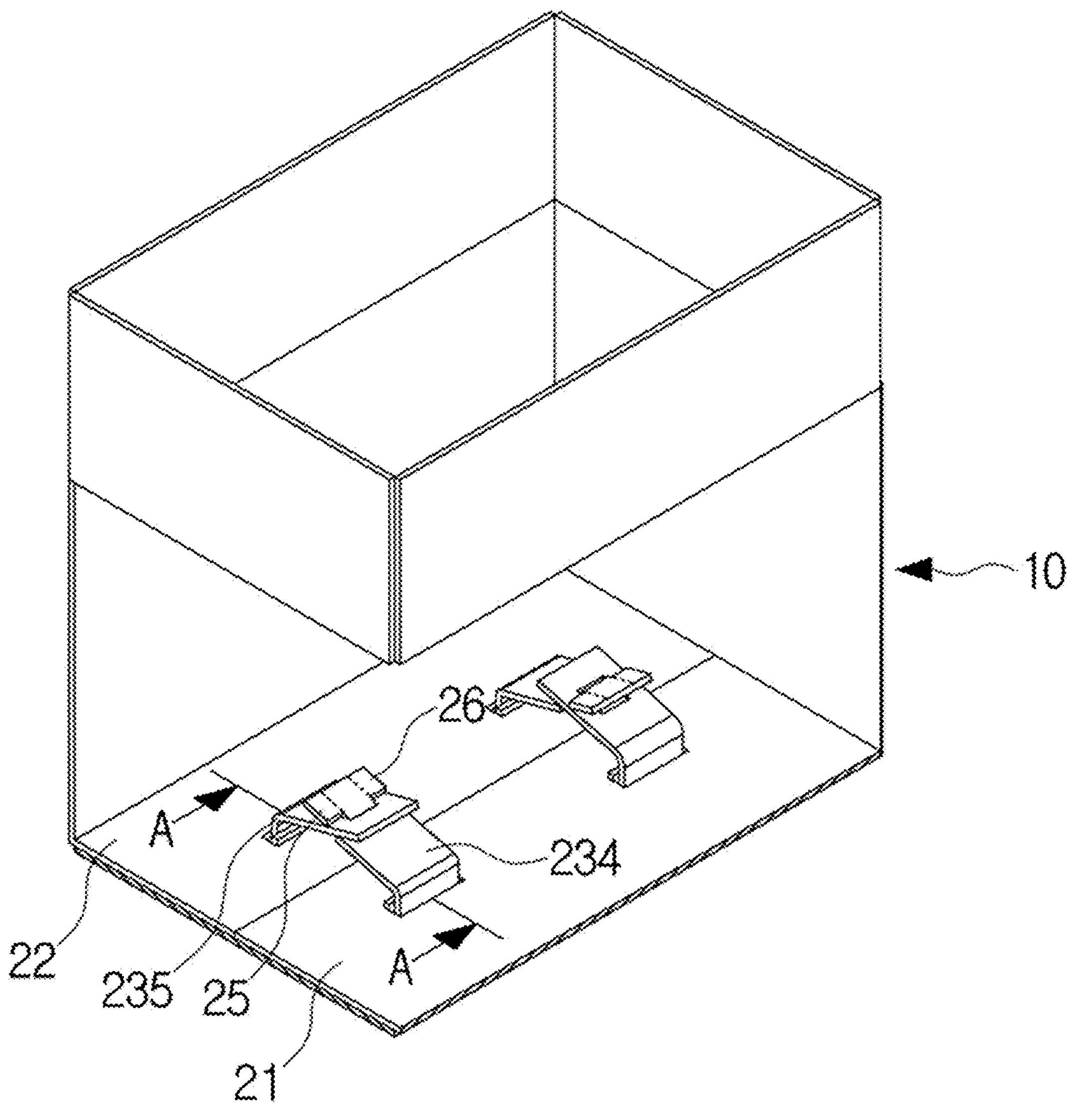


FIG. 15

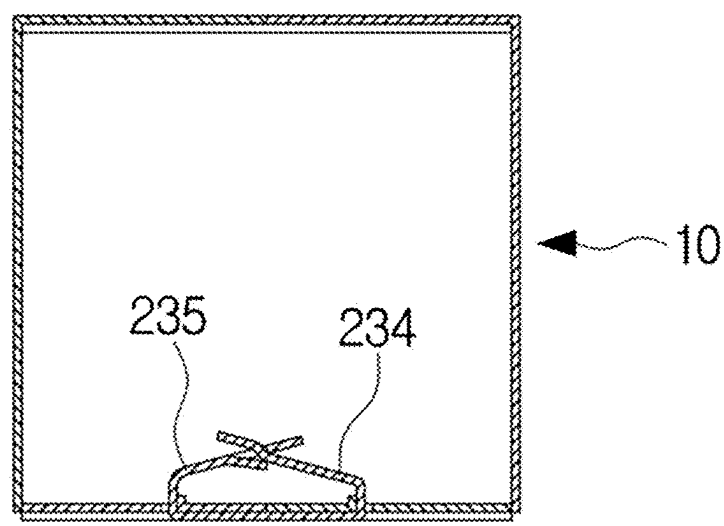


FIG. 16

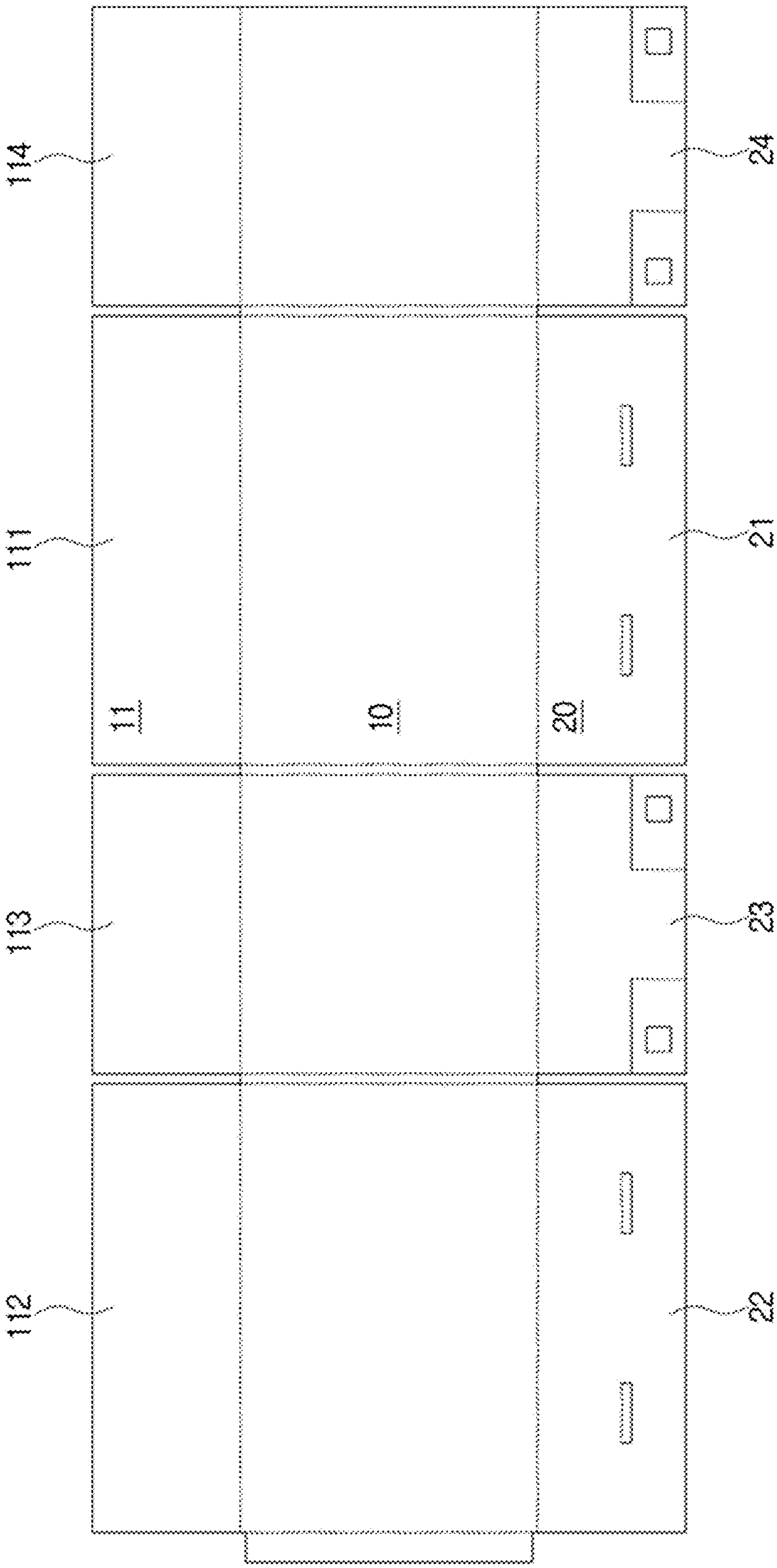


FIG. 17

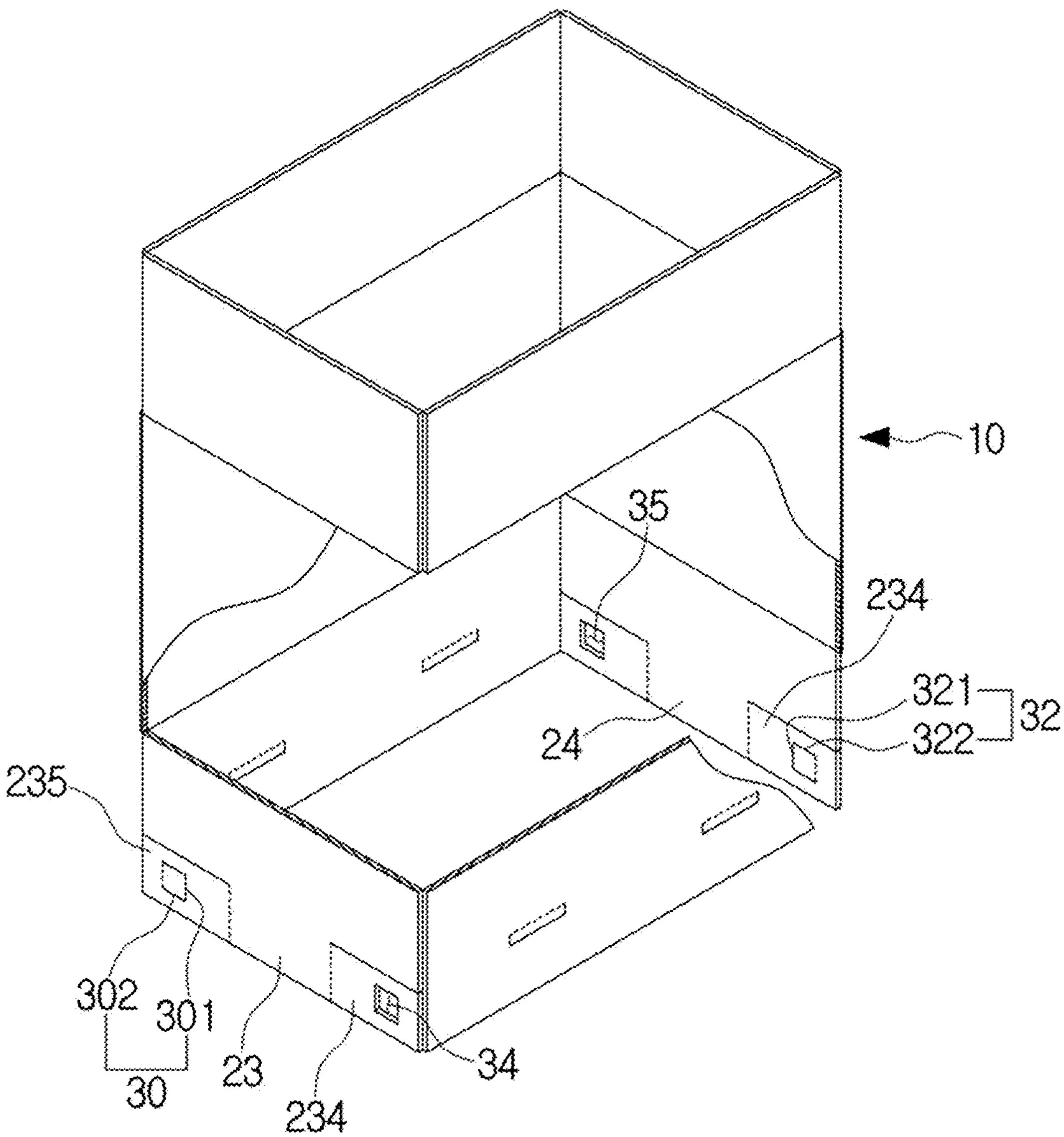


FIG. 18

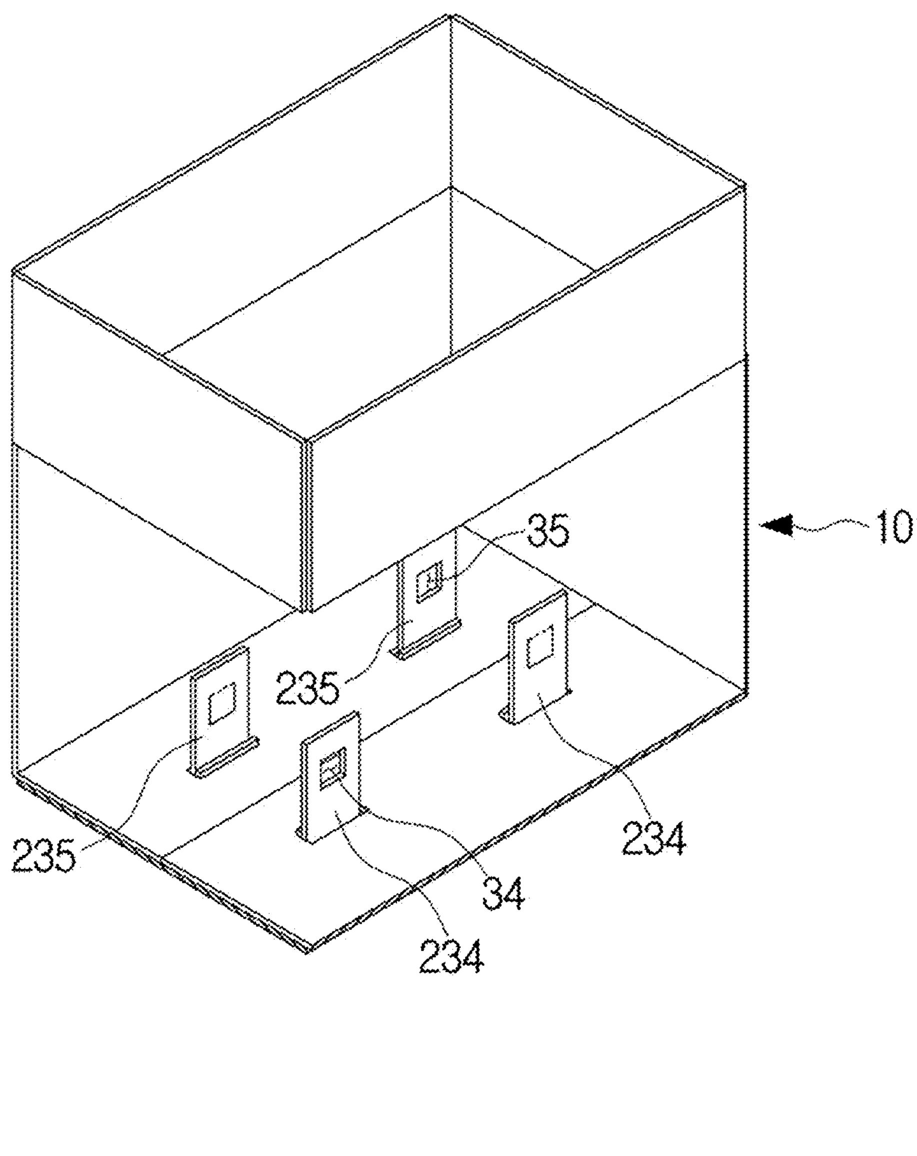


FIG. 19

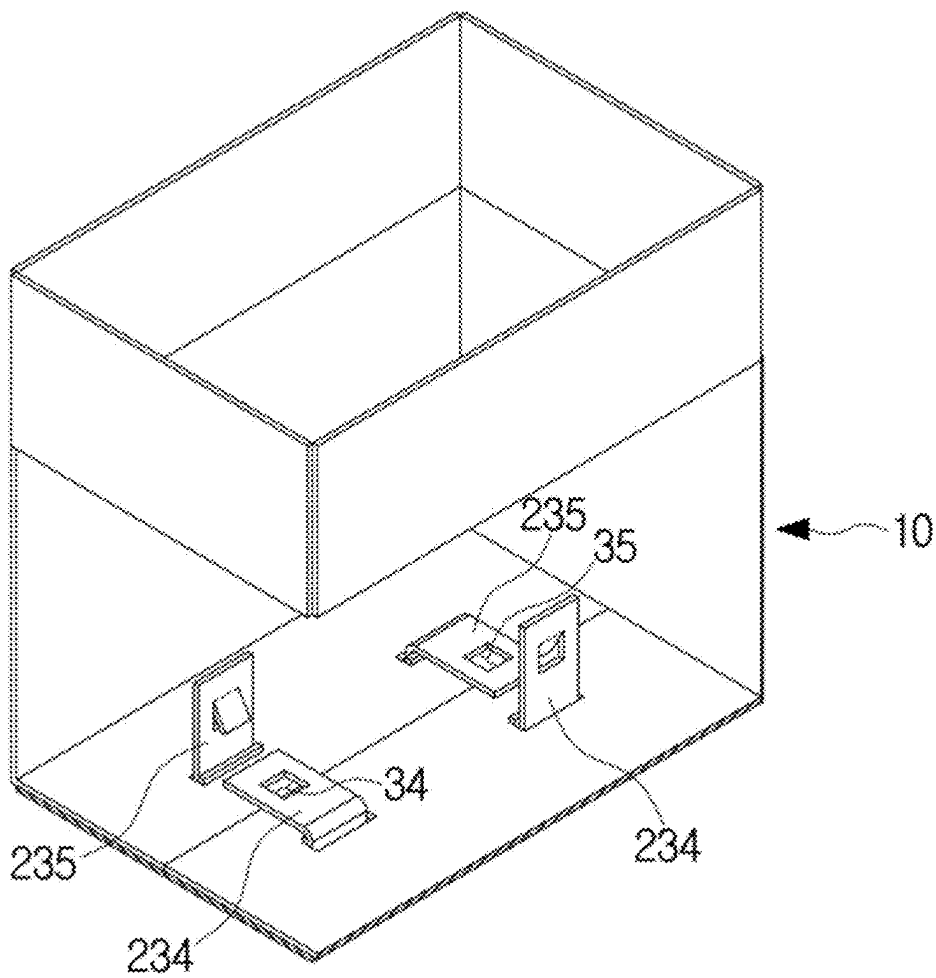


FIG. 20

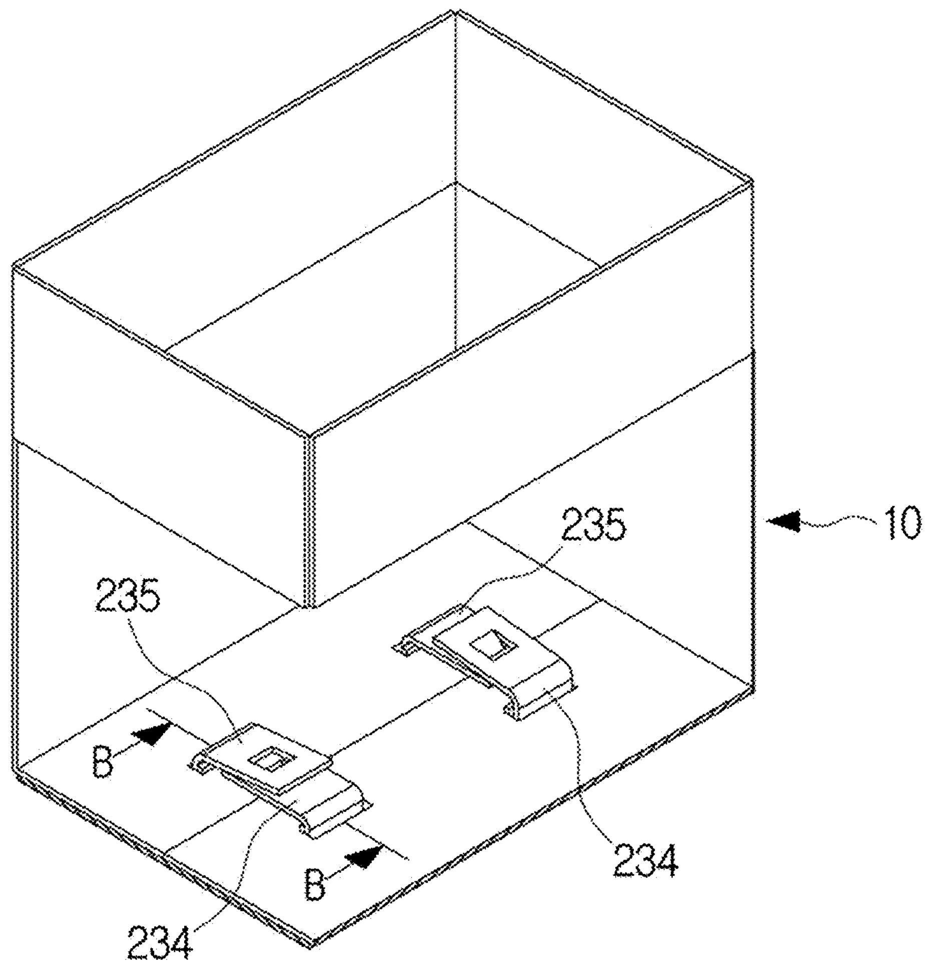
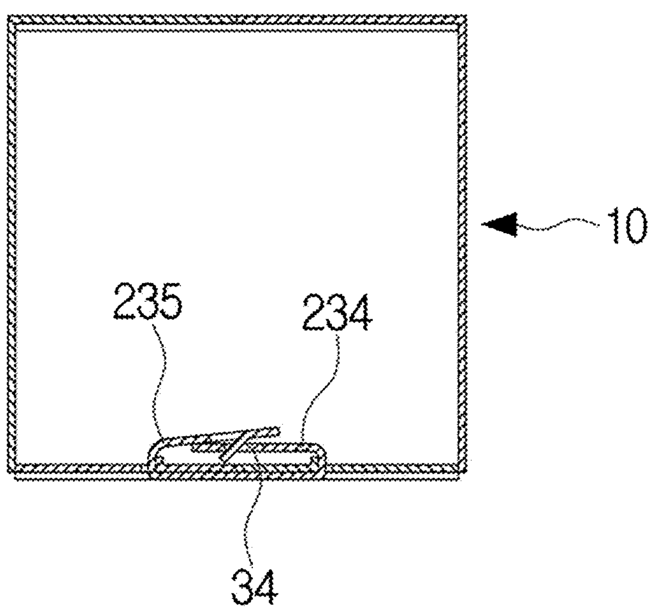


FIG. 21



PACKAGING BOX

TECHNICAL FIELD

[0001] The present invention relates to a packaging box that enables a quadrangular box main body to be assembled without using packaging tape, string, or a separate portable holder clip, can hold and easily transport various kinds of heavy-duty objects, such as daily necessities, using the assembled quadrangular box main body, and does not cause concern of its secure state coming undone due to the heavy-duty objects.

BACKGROUND ART

[0002] Nowadays, due to the enormous amount of plastic waste generated by packaging tape, string, and the like used in self-packaging in supermarkets, supermarkets only provide boxes and do not provide tape or string, making it difficult for consumers to use boxes at self-packaging stations.

[0003] Accordingly, there is a problem that empty boxes generated in supermarkets, which have been conveniently used by consumers, are likely to be discarded without being reused.

[0004] In order to address such a problem, in recent years, Korean Unexamined Patent Application Publication No. 10-2020-0011039 has proposed a fixing method using a protrusion on an attachment surface on the bottom of a packaging box, the fixing method enabling an empty box to be reused by fixing the bottom of the box with a portable holder clip instead of adhesive box packaging tape.

[0005] However, in the case of the above Korean Unexamined Patent Application Publication No. 10-2020-0011039, there are inconveniences that the portable holder clip for fixing the bottom of a box to allow reuse of the empty box has to be separately manufactured and then purchased and owned by a consumer.

DISCLOSURE

Technical Problem

[0006] The present invention has been devised to address the above problems and is directed to providing a packaging box that enables a quadrangular box main body to be assembled without using packaging tape, string, or a separate portable holder clip, can hold and easily transport various kinds of heavy-duty objects, such as daily necessities, using the assembled quadrangular box main body, and does not cause concern of its secure state coming undone due to the heavy-duty objects.

Technical Solution

[0007] The present invention for achieving the above objective provides a packaging box including a quadrangular box main body; and bottom forming parts each formed to extend from a lower end portion of the quadrangular box main body and bent inward to form a bottom, wherein the bottom forming parts include a front side bottom plate and a rear side bottom plate that are formed to extend from a front side of the lower end portion of the quadrangular box main body and a rear side of the lower end portion, respectively, toward a lower portion of the quadrangular box main body to have ends touching each other when bent inward from the quadrangular box main body and that have a one-side slit and an other-side slit formed at one side and the other side, respectively, in a horizontal direction; and a one-side bottom plate and an other-side bottom plate that are formed to extend from one side of the lower end portion of the quadrangular box main body and the other side of the lower end portion, respectively, toward the lower portion of the quadrangular box main body to be bent inward from the quadrangular box main body and that have a front side cutting line and a rear side cutting line formed at a front side and a rear side, respectively, in the horizontal direction and a front side wing part and a rear side wing part respectively formed on a front side portion downward of the front side cutting line and a rear side portion downward of the rear side cutting line, and in a process in which the front side bottom plate and the rear side bottom plate are each bent inward from the quadrangular box main body, and then the one-side bottom plate and the other-side bottom plate are bent inward from the quadrangular box main body, the front side wing part and the rear side wing part are secured inside the quadrangular box main body in a state in which the front side wing part and the rear side wing part pass through the one-side slit and the other-side slit, respectively.

Advantageous Effects

[0008] According to the present invention, a front side wing part and a rear side wing part formed on a one-side bottom plate and an other-side bottom plate can be secured inside a quadrangular box main body in a state in which the front side wing part and the rear side wing part pass through a one-side slit and an other-side slit of a front side bottom plate and a rear side bottom plate, thereby forming a bottom forming part of the quadrangular box main body, and thus the quadrangular box main body can be assembled in a state in which a lower portion of the quadrangular box main body is easily closed without using packaging tape, string, or a separate portable holder clip, various kinds of heavy-duty objects, such as daily necessities, can be held inside the assembled quadrangular box main body and easily transported, and there is no concern about a secure state of the front side wing part and the rear side wing part coming undone due to the heavy-duty objects.

DESCRIPTION OF DRAWINGS

[0009] FIG. **1** is a development view schematically illustrating a packaging box according to one embodiment of the present invention.

[0010] FIGS. **2** to **4** are cutaway perspective views sequentially illustrating a typical process of assembling bottom forming parts and a process of closing an upper portion of a quadrangular box main body using tape.

[0011] FIG. **5** is a perspective view schematically illustrating a state in which the quadrangular box main body is flipped upside down.

[0012] FIG. **6** is a perspective view schematically illustrating a state in which the bottom forming parts of the upside-down quadrangular box main body are unfolded.

[0013] FIG. **7** is a perspective view schematically illustrating a state in which a front side bottom plate and a rear side bottom plate are bent inward from the upside-down quadrangular box main body.

[0014] FIGS. **8** to **10** are perspective views sequentially illustrating a process in which, as a one-side bottom plate and an other-side bottom plate are bent inward from the upside-down quadrangular box main body, a front side wing part of one example and a rear side wing part of one example pass through a one-side slit and an other-side slit.

[0015] FIGS. **11** to **14** are cutaway perspective views sequentially illustrating a process in which the front side wing part of one example and the rear side wing part of one example are secured inside the quadrangular box main body.

[0016] FIG. **15** is a cross-sectional view along A-A of FIG. **14**.

[0017] FIG. **16** is a development view schematically illustrating a packaging box according to another embodiment of the present invention.

[0018] FIG. **17** is a perspective view schematically illustrating a state in which a front side bottom plate, a rear side bottom plate, a one-side bottom plate, and an other-side bottom plate, on which a front side wing part of another example and a rear side wing part of another example are formed, are unfolded.

[0019] FIGS. **18** to **20** are cutaway perspective views sequentially illustrating a process in which the front side wing part of another example and the rear side wing part of another example are secured inside a quadrangular box main body.

[0020] FIG. **21** is a cross-sectional view along B-B of FIG. **20**.

MODES OF THE INVENTION

[0021] Hereinafter, preferred embodiments of the present invention will be described in more detail based on the accompanying drawings. Of course, the scope of rights of the present invention is not limited to the following embodiments, and the present invention may be modified in various ways by those of ordinary skill in the art within the scope not departing from the technical gist of the present invention.

[0022] FIG. **1** is a development view schematically illustrating a packaging box according to one embodiment of the present invention, and FIGS. **2** to **4** are cutaway perspective views sequentially illustrating a typical process of assembling bottom forming parts and a process of closing an upper portion of a quadrangular box main body using tape.

[0023] As shown in FIGS. **1** and **2**, the packaging box according to one embodiment of the present invention mainly includes a quadrangular box main body **10** and a bottom forming part **20**.

[0024] First, an upper portion of the quadrangular box main body **10** may be completely open.

[0025] Alternatively, an upper portion forming part **11** for opening or closing the upper portion of the quadrangular box main body **10** may be formed at the upper portion of the quadrangular box main body **10**.

[0026] For example, as shown in FIG. **2**, the upper portion forming part **11** may mainly include a front side upper plate **111**, a rear side upper plate **112**, a one-side upper plate **113**, and an other-side upper plate **114**.

[0027] The front side upper plate **111** and the rear side upper plate **112** may be formed to vertically extend from a front side of an upper end portion of the quadrangular box main body **10** and a rear side of the upper end portion, respectively, toward the upper portion of the quadrangular box main body **10**.

[0028] For an upper end portion of the front side upper plate **111** and an upper end portion of the rear side upper plate **112** to touch each other, the front side upper plate **111** and the rear side upper plate **112** may each be horizontally bent upward inside the quadrangular box main body **10**.

[0029] The one-side upper plate **113** and the other-side upper plate **114** may be formed to vertically extend from one side of the upper end portion of the quadrangular box main body **10** and the other side of the upper end portion, respectively, toward the upper portion of the quadrangular box main body **10**.

[0030] The one-side upper plate **113** and the other-side upper plate **114** may each be horizontally bent upward inside the quadrangular box main body **10**.

[0031] In a state in which the front side upper plate **111** and the rear side upper plate **112** are each horizontally bent upward inside the quadrangular box main body **10** after the one-side upper plate **113** and the other-side upper plate **114** are each horizontally bent upward inside the quadrangular box main body **10**, as shown in FIG. **4**, packaging tape **3** may be attached at a predetermined length from one side toward the other side of the upper portion forming part **11** at a middle portion of the upper portion forming part **11**.

[0032] Next, as shown in FIG. **2**, the bottom forming part **20** may mainly include a front side bottom plate **21**, a rear side bottom plate **22**, a one-side bottom plate **23**, and an other-side bottom plate **24**.

[0033] The front side bottom plate **21** and the rear side bottom plate **22** may be formed to vertically extend a predetermined length from a front side of a lower end portion of the quadrangular box main body **10** and a rear side of the lower end portion, respectively, toward a lower portion of the quadrangular box main body **10**.

[0034] For a lower end portion of the front side bottom plate **21** and a lower end portion of the rear side bottom plate **22** to touch each other, the front side bottom plate **21** and the rear side bottom plate **22** may each be horizontally bent inward from the quadrangular box main body **10**.

[0035] A one-side slit **211** and an other-side slit **212** horizontally extending a predetermined length to the left and right may be formed from one side toward the other side of the front side bottom plate **21** and from one side toward the other side of the rear side bottom plate **22** at a one-side middle portion and an other-side middle portion of the front side bottom plate **21** and a one-side middle portion and an other-side middle portion of the rear side bottom plate **22**.

[0036] The one-side bottom plate **23** and the other-side bottom plate **24** may be formed to vertically extend a predetermined length from one side of the lower end portion of the quadrangular box main body **10** and the other side of the lower end portion, respectively, toward the lower portion of the quadrangular box main body **10** and may each be horizontally bent inward from the quadrangular box main body **10**.

[0037] A front side cutting line **231** and a rear side cutting line **232** that horizontally extend to a predetermined height in a front-rear direction of the one-side bottom plate **23** and in a front-rear direction of the other-side bottom plate **24** may be formed at a front side and a rear side of the one-side bottom plate **23** and a front side and a rear side of the other-side bottom plate **24**.

[0038] In addition, a front side wing part **234** may be formed at a front side portion of the one-side bottom plate **23** and a front side portion of the other-side bottom plate **24** downward from the front side cutting line **231**.

[0039] A rear side wing part **235** may be formed at a rear side portion of the one-side bottom plate **23** and a rear side portion of the other-side bottom plate **24** downward from the rear side cutting line **232**.

[0040] For example, first, as shown in FIG. **3**, the one-side bottom plate **23** and the other-side bottom plate **24** are each horizontally bent inward from the quadrangular box main body **10**, and then, as shown in FIG. **4**, the front side bottom plate **21** and the rear side bottom plate **22** may each be horizontally bent inward from the quadrangular box main body **10**.

[0041] FIG. **5** is a perspective view schematically illustrating a state in which the quadrangular box main body is flipped upside down, and FIG. **6** is a perspective view schematically illustrating a state in which the bottom forming parts of the upside-down quadrangular box main body are unfolded.

[0042] Meanwhile, in particular, in order to enable the quadrangular box main body **10** to be assembled in a state in which the lower portion of the quadrangular box main body **10** is easily closed without using packaging tape, string, or a separate portable holder clip, first, as shown in FIGS. **5** and **6**, the front side bottom plate **21**, the rear side bottom plate **22**, the one-side bottom plate **23**, and the other-side bottom plate **24**, each horizontally bent inward from the quadrangular box main body **10**, may be unfolded back to a vertical state.

[0043] FIG. **7** is a perspective view schematically illustrating a state in which a front side bottom plate and a rear side bottom plate are bent inward from the upside-down quadrangular box main body.

[0044] In this state, as shown in FIG. **7**, the front side bottom plate **21** and the rear side bottom plate **22** may each be horizontally bent inward from the quadrangular box main body **10**.

[0045] FIGS. **8** to **10** are perspective views sequentially illustrating a process in which, as a one-side bottom plate and an other-side bottom plate are bent inward from the upside-down quadrangular box main body, a front side wing part of one example and a rear side wing part of one example pass through a one-side slit and an other-side slit.

[0046] In addition, as shown in FIGS. **8** to **10**, in a process in which the one-side bottom plate **23** and the other-side bottom plate **24** are horizontally bent inward from the quadrangular box main body **10**, the front side wing part **234** and the rear side wing part **235** may vertically pass through the one-side slit **211** and the other-side slit **212**, respectively, while bent toward the one-side slit **211** and the other-side slit **212**, respectively.

[0047] Here, for the front side wing part **234** and the rear side wing part **235** to more easily and stably vertically pass through the one-side slit **211** and the other-side slit **212**, respectively, preferably, a left-right width (W_1 of FIG. **1**) of the one-side slit **211** and a left-right width (W_1 of FIG. **1**) of the other-side slit **212** may be the same as an up-down width (W_2 of FIG. **1**) of the front side wing part **234** and an up-down width (W_2 of FIG. **1**) of the rear side wing part **235**.

[0048] In addition, in order to efficiently prevent some daily necessities stored inside the quadrangular box main body **10** from passing through the one-side slit **211** and the other-side slit **212** and falling outward through the lower portion of the quadrangular box main body **10**, as shown in FIG. **2**, a one-side cutting line **213** and an other-side cutting line **214**, each having a rectangular shape with an open top, may be formed at a one-side middle portion and an other-side middle portion of the front side bottom plate **21** and a one-side middle portion and an other-side middle portion of the rear side bottom plate **22**.

[0049] For example, the one-side cutting line **213** and the other-side cutting line **214** may each be mainly formed of a first horizontal cutting line **201** and a first vertical cutting line **202**.

[0050] The first horizontal cutting line **201** may be formed at a predetermined height while horizontally extending to the left and right from one side toward the other side of the front side bottom plate **21** and from one side toward the other side of the rear side bottom plate **22** at a one-side middle portion and an other-side middle portion of the front side bottom plate **21** and a one-side middle portion and an other-side middle portion of the rear side bottom plate **22**.

[0051] The first vertical cutting line **202** may be formed to vertically extend a predetermined length upward from the first horizontal cutting line **201** at each of one side and the other side of the first horizontal cutting line **201**.

[0052] A portion of the front side bottom plate **21** and a portion of the rear side bottom plate **22** in a direction inward from the one-side cutting line **213** and a direction inward from the other-side cutting line **214** may close the one-side slit **211** and the other-side slit **212** at ordinary times.

[0053] In addition, as the portion of the front side bottom plate **21** and the portion of the rear side bottom plate **22** in the direction inward from the one-side cutting line **213** and the direction inward from the other-side cutting line **214** are folded inward from the quadrangular box main body **10**, the one-side slit **211** and the other-side slit **212**, through which the front side wing part **234** and the rear side wing part **235** are able to vertically pass, may be formed.

[0054] Next, for the front side wing part **234** and the rear side wing part **235** to be easily bent toward the one-side slit **211** and the other-side slit **212**, respectively, as shown in FIGS. **8** to **10**, preferably, a perforated line **29** extending in an up-down direction of the front side wing part **234** and an up-down direction of the rear side wing part **235** may be formed at each of a rear side of the front side wing part **234** and a front side of the rear side wing part **235** as shown in FIGS. **2** and **7**.

[0055] Next, for example, as shown in FIG. **2**, based on the one-side bottom plate **23**, an upper cutting line **203** in the vertical direction and a lower cutting line **204** in the vertical direction may be formed at an upper portion of a front side of the front side wing part **234** and a lower portion of the front side.

[0056] An upper securing wing **25** and a lower securing wing **26** that are foldable to be orthogonal to the front side wing part **234** may be formed at each of a portion of the front side wing part **234** that is forward from the upper cutting line **203** and a portion of the front side wing part **234** that is forward from the lower cutting line **204**.

[0057] In addition, a through-hole **236** through which the upper securing wing **25** and the lower securing wing **26**, which are in a folded state, pass may be formed at a rear side of the rear side wing part **235**.

[0058] In another example, the upper securing wing **25** and the lower securing wing **26** may be formed at portions of the rear side wing part **235**, and a through-hole **237** may be formed at a portion of the front side wing part **234**.

[0059] For example, as shown in FIG. **2**, based on the other-side bottom plate **24**, the upper cutting line **203** in the vertical direction and the lower cutting line **204** in the vertical direction may be formed at an upper portion of the front side of the rear side wing part **235** and a lower portion of the front side.

[0060] The upper securing wing **25** and the lower securing wing **26** that are foldable to be orthogonal to the rear side wing part **235** may be formed at each of a portion of the rear side wing part **235** that is rearward from the upper cutting line **203** and a portion of the rear side wing part **235** that is rearward from the lower cutting line **204**.

[0061] The through-hole **237** through which the upper securing wing and the lower securing wing, which are in a folded state, pass may be formed at the front side of the front side wing part **234**.

[0062] Next, for each of the upper securing wing **25** and the lower securing wing **26** to be more smoothly foldable, as shown in FIG. **2**, a perforated line **27** extending a predetermined length to the left and right from one side toward the other side of the upper securing wing **25** and a perforated line **27** extending a predetermined length to the left and right from one side toward the other side of the lower securing wing **26** may be formed at a lower portion of the upper securing wing **25** and an upper portion of the lower securing wing **26**, respectively.

[0063] Next, in order to efficiently prevent some daily necessities stored inside the quadrangular box main body **10** from passing through the through-holes **236** and **237** and falling outward through the lower portion of the quadrangular box main body **10**, first, as shown in FIG. **2**, based on the one-side bottom plate **23**, a rear side auxiliary cutting line **30** having a rectangular shape with an open right side may be formed at the rear side of the rear side wing part **235**.

[0064] For example, the rear side auxiliary cutting line **30** may be mainly formed of a first vertical auxiliary cutting line **301** and a first horizontal auxiliary cutting line **302**.

[0065] The first vertical auxiliary cutting line **301** may be formed to extend a predetermined length in the up-down direction of the rear side wing part **235** from a middle portion of the rear side of the rear side wing part **235**.

[0066] The first horizontal auxiliary cutting line **302** may be formed to horizontally extend a predetermined length in a forward direction of the first vertical auxiliary cutting line **301** from each of an upper portion and a lower portion of the first vertical auxiliary cutting line **301**

[0067] A portion of the rear side wing part **235** that is inward of the rear side auxiliary cutting line **30** may close the through-hole **236** at ordinary times.

[0068] As the portion of the rear side wing part **235** that is inward of the rear side auxiliary cutting line **30** is folded to be orthogonal to the rear side auxiliary cutting line **30**, the through-hole **236** may be formed inside of the rear side auxiliary cutting line **30**.

[0069] In addition, as shown in FIG. **2**, based on the other-side bottom plate **24**, a front side auxiliary cutting line **32** having a rectangular shape with an open left side may be formed at the front side of the front side wing part **234**.

[0070] The front side auxiliary cutting line **32** may be formed of a first vertical auxiliary cutting line **321** and a first horizontal auxiliary cutting line **322**.

[0071] The first vertical auxiliary cutting line **321** may be formed to extend a predetermined length in the up-down direction of the front side wing part **234** from a middle portion of the front side of the front side wing part **234**.

[0072] The first horizontal auxiliary cutting line **322** may be formed to horizontally extend a predetermined length in a rearward direction of the first vertical auxiliary cutting line **321** from each of an upper portion and a lower portion of the first vertical auxiliary cutting line **321**.

[0073] A portion of the front side wing part **234** that is inward of the front side auxiliary cutting line **32** may close the through-hole **237** at ordinary times.

[0074] As the portion of the front side wing part **234** that is inward of the front side auxiliary cutting line **32** is folded to be orthogonal to the front side auxiliary cutting line **32**, the through-hole **237** may be formed inside of the front side auxiliary cutting line **32**.

[0075] FIGS. **11** to **14** are cutaway perspective views sequentially illustrating a process in which the front side wing part of one example and the rear side wing part of one example are secured inside the quadrangular box main body, and FIG. **15** is a cross-sectional view along A-A of FIG. **14**.

[0076] As shown in FIGS. **11** to **14**, the upper securing wing **25** and the lower securing wing **26** located inside the quadrangular box main body **10** may, while folded to be orthogonal to the front side wing part **234** and the rear side wing part **235**, respectively, pass through the through-holes **236** and **237**, respectively, and be unfolded back to a horizontal state.

[0077] In this way, as shown in FIGS. **14** and **15**, the front side wing part **234** and the rear side wing part **235**, which vertically pass through the one-side slit **211** and the other-side slit **212**, respectively, may be secured inside the quadrangular box main body **10** and form the bottom forming part **20** of the quadrangular box main body **10**.

[0078] Various kinds of heavy-duty objects, such as daily necessities purchased from supermarkets or the like, may be held in the assembled quadrangular box main body **10** and easily transported, and there is no concern about a secure state of the front side wing part **234** and the rear side wing part **235** coming undone due to the heavy-duty objects.

[0079] FIG. **16** is a development view schematically illustrating a packaging box according to another embodiment of the present invention, and FIG. **17** is a perspective view schematically illustrating a state in which a front side bottom plate, a rear side bottom plate, and a one-side bottom plate and an other-side bottom plate, on which a front side wing part of another example and a rear side wing part of another example are formed, are unfolded.

[0080] Next, as shown in FIGS. **16** and **17**, a packaging box according to another embodiment of the present invention includes the quadrangular box main body **10** and the bottom forming part **20** like the packaging box according to one embodiment.

[0081] Yet, as shown in FIG. **17**, in one example, based on the one-side bottom plate **23**, a through-hole **34** may be formed at the middle portion of the front side of the front side wing part **234**, and the rear side auxiliary cutting line **30** may be formed at the rear side of the rear side wing part **235**.

[0082] The rear side auxiliary cutting line **30** may be formed of the first vertical auxiliary cutting line **301** and the first horizontal auxiliary cutting line **302**.

[0083] The first vertical auxiliary cutting line **301** may be formed to extend a predetermined length in the up-down direction of the rear side wing part **235** from a middle portion of the rear side wing part **235**.

[0084] The first horizontal auxiliary cutting line **302** may be formed to extend a predetermined length in a rearward direction of the first vertical auxiliary cutting line **301** from each of the upper portion and the lower portion of the first vertical auxiliary cutting line **301**.

[0085] In another example, a through-hole **35** may be formed at the middle portion of the rear side of the rear side wing part **235**, and the front side auxiliary cutting line **32** may be formed at the front side wing part **234**.

[0086] For example, as shown in FIG. **17**, based on the other-side bottom plate **24**, the front side auxiliary cutting line **32** may be formed of the first vertical auxiliary cutting line **321** and the first horizontal auxiliary cutting line **322** as shown in FIG. **17**.

[0087] The first vertical auxiliary cutting line **321** may be formed to extend a predetermined length in the up-down direction of the front side wing part **234** from a middle portion of the front side wing part **234**.

[0088] The first horizontal auxiliary cutting line **322** may be formed to extend a predetermined length in a forward direction of the first vertical auxiliary cutting line **321** from each of the upper portion and the lower portion of the first vertical auxiliary cutting line **321**.

[0089] FIGS. **18** to **20** are cutaway perspective views sequentially illustrating a process in which the front side wing part of another example and the rear side wing part of another example are secured inside a quadrangular box main body, and FIG. **21** is a cross-sectional view along B-B of FIG. **20**.

[0090] As shown in FIGS. **18** to **21**, the front side wing part **234** and the rear side wing part **235** may be folded while overlapping each other so that a portion of the rear side wing part **235** that is inward of the rear side auxiliary cutting line **30** is secured after passing through the through-hole **34** of the front side wing part **234** and being caught in and fixed to the through-hole **34** and a portion of the front side wing part **234** that is inward of the front side auxiliary cutting line **32** is secured after passing through the through-hole **35** of the rear side wing part **235** and being caught in and fixed to the through-hole **35**.

INDUSTRIAL APPLICABILITY

[0091] According to the present invention, a front side wing part and a rear side wing part formed on a one-side bottom plate and an other-side bottom plate can be secured inside a quadrangular box main body in a state in which the front side wing part and the rear side wing part pass through a one-side slit and an other-side slit of a front side bottom plate and a rear side bottom plate, thereby forming a bottom forming part of the quadrangular box main body, and thus the quadrangular box main body can be assembled in a state in which a lower portion of the quadrangular box main body is easily closed without using packaging tape, string, or a separate portable holder clip, various kinds of heavy-duty objects, such as daily necessities, can be held inside the assembled quadrangular box main body and easily transported, and there is no concern about a secure state of the front side wing part and the rear side wing part coming undone due to the heavy-duty objects.

1. A packaging box comprising:

a quadrangular box main body; and

bottom forming parts each formed to extend from a lower end portion of the quadrangular box main body and bent inward to form a bottom,

wherein the bottom forming parts include a front side bottom plate and a rear side bottom plate that are formed to extend from a front side of the lower end portion of the quadrangular box main body and a rear side of the lower end portion, respectively, toward a lower portion of the quadrangular box main body to have ends touching each other when bent inward from the quadrangular box main body and that have a one-side slit and an other-side slit formed at one side and the other side, respectively, in a horizontal direction; and a one-side bottom plate and an other-side bottom plate that are formed to extend from one side of the lower end portion of the quadrangular box main body and the other side of the lower end portion, respectively, toward the lower portion of the quadrangular box main body to be bent inward from the quadrangular box main body and that have a front side cutting line and a rear side cutting line formed at a front side and a rear side, respectively, in the horizontal direction and a front side wing part and a rear side wing part respectively formed on a front side portion downward of the front side cutting line and a rear side portion downward of the rear side cutting line, and

in a process in which the front side bottom plate and the rear side bottom plate are each bent inward from the quadrangular box main body, and then the one-side bottom plate and the other-side bottom plate are bent inward from the quadrangular box main body, the front side wing part and the rear side wing part are secured inside the quadrangular box main body in a state in which the front side wing part and the rear side wing part pass through the one-side slit and the other-side slit, respectively.

2. The packaging box of claim **1**, wherein a left-right width of the one-side slit and a left-right width of the other-side slit of the front side bottom plate and the rear side bottom plate are the same as an up-down width of the front side wing part and an up-down width of the rear side wing part of the one-side bottom plate and the other-side bottom plate.

3. The packaging box of claim **1**, wherein:

a one-side cutting line and an other-side cutting line, each formed of a first horizontal cutting line and a first vertical cutting line extending a predetermined length upward from the first horizontal cutting line from each of one side and the other side of the first horizontal cutting line, are formed at one side and the other side of the front side bottom plate and one side and the other side of the rear side bottom plate; and

as a portion of the front side bottom plate and a portion of the rear side bottom plate in a direction inward from the one-side cutting line and a direction inward from the other-side cutting line are folded inward from the quadrangular box main body, the one-side slit and the other-side slit are formed inside of the one-side cutting line and inside of the other-side cutting line, respectively.

4. The packaging box of claim **1**, wherein a perforated line extending in an up-down direction of the front side wing part and an up-down direction of the rear side wing part is formed at each of a rear side of the front side wing part and a front side of the rear side wing part.

5. The packaging box of claim **1**, wherein:

an upper cutting line in a vertical direction and a lower cutting line in the vertical direction are formed at an upper portion of a front side of the front side wing part and a lower portion of the front side;

an upper securing wing and a lower securing wing that are foldable to be orthogonal to the front side wing part are formed at each of a portion of the front side wing part that is forward from the upper cutting line and a portion of the front side wing part that is forward from the lower cutting line; and

a through-hole through which the upper securing wing and the lower securing wing, which are in a folded state, pass is formed at a rear side of the rear side wing part.

6. The packaging box of claim **1**, wherein:

an upper cutting line in a vertical direction and a lower cutting line in the vertical direction are formed at an upper portion of a front side of the rear side wing part and a lower portion of the front side;

an upper securing wing and a lower securing wing that are foldable to be orthogonal to the rear side wing part are formed at each of a portion of the rear side wing part that is rearward from the upper cutting line and a portion of the rear side wing part that is rearward from the lower cutting line; and

a through-hole through which the upper securing wing and the lower securing wing, which are in a folded state, pass is formed at a front side of the front side wing part.

7. The packaging box of claim **5**, wherein a perforated line extending in a left-right direction of the upper securing wing and a left-right direction of the lower securing wing is formed at a lower portion of the upper securing wing and an upper portion of the lower securing wing.

8. The packaging box of claim **5**, wherein:

a rear side auxiliary cutting line, formed of a first vertical auxiliary cutting line and a first horizontal auxiliary cutting line extending a predetermined length forward from the first vertical auxiliary cutting line from each of an upper portion and a lower portion of the first vertical auxiliary cutting line, is formed at the rear side of the rear side wing part; and

as a portion of the rear side wing part that is in a direction inward from the rear side auxiliary cutting line is folded to be orthogonal to the rear side auxiliary cutting line, the through-hole is formed inside of the rear side auxiliary cutting line.

9. The packaging box of claim **6**, wherein:

a front side auxiliary cutting line, formed of a first vertical auxiliary cutting line and a first horizontal auxiliary cutting line extending a predetermined length rearward from the first vertical auxiliary cutting line from each of an upper portion and a lower portion of the first vertical auxiliary cutting line, is formed at the front side of the front side wing part; and

as a portion of the front side wing part that is in a direction inward from the front side auxiliary cutting line is folded to be orthogonal to the front side auxiliary cutting line, the through-hole is formed inside of the front side auxiliary cutting line.

10. The packaging box of claim **1**, wherein:

a through-hole is formed at a front side of the front side wing part;

a rear side auxiliary cutting line, formed of a first vertical auxiliary cutting line and a first horizontal auxiliary cutting line extending a predetermined length rearward from the first vertical auxiliary cutting line from each of an upper portion and a lower portion of the first vertical auxiliary cutting line, is formed at a rear side of the rear side wing part; and

a portion of the rear side wing part that is in a direction inward from the rear side auxiliary cutting line is secured after passing through the through-hole of the front side wing part and being caught in and fixed to the through-hole.

11. The packaging box of claim **1**, wherein:

a through-hole is formed at a rear side of the rear side wing part;

a front side auxiliary cutting line, formed of a first vertical auxiliary cutting line and a first horizontal auxiliary cutting line extending a predetermined length forward from the first vertical auxiliary cutting line from each of an upper portion and a lower portion of the first vertical auxiliary cutting line, is formed at a front side of the front side wing part; and

a portion of the front side wing part that is in a direction inward from the front side auxiliary cutting line is secured after passing through the through-hole of the rear side wing part and being caught in and fixed to the through-hole.

12. The packaging box of claim **6**, wherein a perforated line extending in a left-right direction of the upper securing wing and a left-right direction of the lower securing wing is formed at a lower portion of the upper securing wing and an upper portion of the lower securing wing.

* * * * *